

# Affine Frame Registration and Coordinate Products

Stage C0A2: physical, translation-registered, and reference-material views

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## Scope and dependencies

Stage C0A2 consumes the source-field and periodic lattice-gauge contract from C0A1. It constructs analysis-specific affine maps but does not migrate MSD, VACF, density, plotting, or structural consumers; that work belongs to C0B and C0A3.

The package uses row vectors. For frame  $t$ ,

$$q_{i,t} = x_{i,t}M_t + b_t, \quad G_t = H_tM_t,$$

where  $H_t$  is the C0A1 gauge-reconciled source cell.

## Policies

### Physical

$$M_t = I, \quad b_t = 0, \quad G_t = H_t.$$

### Translation registered

$$M_t = I, \quad b_t = -\tau_t, \quad G_t = H_t.$$

### Reference material

For a full-rank, fully periodic reference cell  $H_{\text{ref}}$ ,

$$M_t = H_t^{-1}H_{\text{ref}}, \quad G_t = H_{\text{ref}}.$$

The policy may be composed with the same matched-reference translation gauge. Reference material registration preserves material/fractional internal motion; it is not a physical bond-length coordinate system.

## Persistent reference and translation gauge

A reference set is a deterministic tuple of atom indices with center-of-geometry, center-of-mass, or explicit positive weights. Reference coordinates are taken from one declared source frame after lattice-gauge reconciliation and mapped into each frame’s pre-translation registered cell.

For residuals  $d_{j,t} = \tilde{q}_{j,t} - q_{j,\text{ref},t}$ , solve on the registered torus

$$\tau_t = \arg \min_{\tau} \sum_j w_j \|\text{MIC}_{G_t,R}(d_{j,t} - \tau)\|_R^2.$$

The implementation forms deterministic candidates by repeatedly lifting all residuals around a seed, taking their weighted mean, and resolving each lift with the certified closest-image solver. Candidate minima are reduced modulo the lattice and deduplicated. The selected result records weighted RMS, maximum residual, competing-minimum separation, closest-image ambiguity, convergence, and reference provenance. This matched-displacement solver is package-specific; it is not a wrapped-coordinate center average.

## Temporal translation-branch lift

The torus representative obeys

$$\tau_t^{\text{lift}} = \tau_t + n_t G_t, \quad n_t \in \mathbb{Z}^3.$$

For a trajectory, each continuous segment chooses the current lattice branch closest to the previous lifted translation under the fit metric. Declared segment starts reset the lift. A closest-image tie fails closed. Independent ensembles receive torus representatives only and do not acquire invented temporal continuity.

## Coordinate products

Every result exposes:

- `registered_unwrapped_cartesian;`
- `registered_wrapped_fractional;`
- `registered_image_shifts;`
- `registered_cells;`

- framewise  $M_t$ , torus translation, and lifted translation.

They obey

$$q = (s + k)G,$$

where  $s$  is wrapped fractional position and  $k$  is the integer image shift. Position round trips, affine inverse round trips, and cell identities are validated before success.

## Displacements and forces

Same-frame displacement transforms as

$$\Delta q = \Delta x M_t.$$

For an externally defined affine map, force is a covector:

$$F_q = F_x M_t^{-\top}.$$

The implementation verifies numerical work invariance. A matched translation fitted from the same atoms is not promoted automatically to an exact generalized force. Exact translation-relative force status requires an explicitly declared target set disjoint from the reference set; otherwise the result is diagnostic while the independent PMF admissibility status from C0A1 is retained.

## Failure behavior

The stage fails closed for:

- a source contract not bound to the collection;
- missing reference cell for `reference_material`;
- partial periodicity for matched torus translation or reference-material mapping;
- singular/incompatible affine maps;
- invalid atom indices or nonpositive weights;
- nonconstant reference set;
- unresolved closest-image or branch-lift ties;
- fixed-domain requests whose registered cells are not constant within tolerance;
- coordinate, cell, or force-work validation failures.

## Acceptance requirements

- $G_t = H_t M_t$  for every frame;
- reference-material mapping yields the declared fixed cell;
- a common framework translation is removed across a periodic boundary;
- branch lifting preserves continuous unwrapped trajectories and respects reset boundaries;
- wrapped coordinates and image shifts exactly reconstruct unwrapped registered positions;
- affine forward/inverse round trips pass;
- covector-transformed forces preserve  $F \cdot \Delta x$ ;
- C0A1 unimodular basis reconciliation does not change registered physical positions;
- deterministic serialization preserves signatures.